Program : **Diploma in Civil and Environmental Engineering**	
Course Code : **3356**	Course Title: **Advanced Survey Practical Lab**
Semester : **3**	Credits: **1.5**
Course Category: **Program core**	
Periods per week: **3(L:0 T:0 P:3)**	Periods per semester:**45**

Course Objectives:

- To enable the students to use theodolite for angle measurement and traversing.
- To develop skills in the students to perform tacheometric surveying to find heights and distances.
- To impart the training in total station and advanced surveying instruments.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of Basic Mathematics		Engineering Mathematics	1 & 2
Basic Surveying		Basic Surveying	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Measure horizontal and vertical angles using theodolite and perform traversing using theodolite and prepare plans	14	Applying
CO2	Determine heights and distances using tacheometry	9	Applying
CO3	Apply Total station survey in field works	12	Applying
CO4	Employ GPS survey in field works	6	Applying
	Lab Tests	4	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Measure horizontal and vertical angles using theodolite and perform theodolite traversing to find out the area of a given plot.**		
M1.01	Measure horizontal angles by repetition method	3	Applying
M1.02	Measure horizontal angles by reiteration method	3	Applying
M1.03	Determine the difference in elevation and horizontal distance between two stations by measuring vertical and horizontal angles from a single station	3	Applying
M1.04	Perform theodolite traversing and record observation.	3	Applying
M1.05	Compute the area of the traverse using Gale's traverse table.	2	Applying
CO2	**Determine heights and distances using tacheometry**		
M2.01	Determine the values of stadia constants of the given instrument.	3	Applying
M2.02	Determine the horizontal distance and difference in elevation between two stations by stadia tacheometry.	3	Applying
M2.03	Determine the horizontal distance and difference in elevation between two stations by tangential tacheometry.	3	Applying
	Lab Test 1	2	

CO3	**Apply Total station survey in field works**		
M3.01	Perform the temporary setting of total station	3	Applying
M3.02	Determine gradient, difference in height, distance etc. between inaccessible points using total station	3	Applying
M3.03	Prepare site plan by traversing with total station	6	Applying
CO4	**Employ GPS survey in field works**		
M4.01	Utilize GPS to find out coordinates (longitude and latitude) of a station	6	Applying
	Lab Test 2	2	

Text / Reference:

T/R	Book Title/Author
T1	Punmia, B.C., Jain, A.K and Jain, A.K. Surveying vol I and II , Laxmi Publications, New Delhi.
R2	Basak, N. N. Surveying and Levelling, McGraw Hill Education, New Delhi.
R3	Kanetkar, T.P and Kulkarni, S.V. Surveying and Levelling volume I and II, Pune Vidyarthi Griha Prakashan
R4	Duggal, S.K. Surveying volume I and II, McGraw Hill Education, New Delhi
R5	Saikia ,M.D., Das, B.M. AND Das, M.M. Surveying, PHI Learning, New Delhi
R6	Subramanian, R. Fundamentals of Surveying and Levelling, Oxford University Press, New Delhi
R7	De, Alak. Plane Surveying, S Chand Publications, New Delhi.
R8	Venkatramaiah, C. Textbook of Surveying, Universities Press, Hyderabad.
R9	Anderson, J M and Mikhail, E M. Surveying theory and practice, McGraw Hill Education, Noida.
R10	Hoffman Wellenhof, B , Lichtenegger, H and Collins, J . Glabal Positioning System- Theory and Practice, Springer Vienna

Online resources:

Sl.No.	Website Link
1	http://www.vlab.co.in/ba-nptel-labs-civil-engineering
2	https://nptel.ac.in/courses
3	https://archive.nptel.ac.in/courses/105/107/105107157//